PROJECT

STUDY ON THE USE OF FUEL CELLS AND HYDROGEN: SUMMARY / APPLICATION CASE

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CONTENT

- SUMMARY
- MOTIVATION
- DRIVER
- WHY H2-PROPULSION?
- CHALLENGES FOR FUEL CELLS IN SHIPPING
- PRELIMINARY DEFINITION OF A HYDROGEN-POWERED SHIP PROPULSION
- APPLICATION CASE
- INFRASTRUCTURE
- CONCLUSION
- REFERENCES
- ANNEX

SUMMARY

- **Summary of the scope of work**
- **Reports/Documents delivered:**
  - StudyOnTheUseOfFuelCellsAndHydrogenAnAlternativePropulsionSystem_200116.pdf
  - StudyOnTheUseOfFuelCellsAndHydrogen_AlternativeFuelsInsight_Provider_200210.pdf
  - StudyOnTheUseOfFuelCellsAndHydrogen_FuelCells_Provider_200214.pdf
  - StudyOnTheUseOfFuelCellsAndHydrogen_HydrogenStorageSelection_200220.pdf
  - Berechnung Antriebskonzept-WIP04_200222.xlsx

MOTIVATION

- Searching an alternative propulsion solution for an innovative research vessel based on hydrogen fuel with a zero-emission target
- Improvement of ship energy efficiency
- Use of alternative fuels
- Zero-Emission Ship is possible:
  - If Hydrogen is generated using Renewable energy (i.e. electrolysis), nuclear power or natural gas with carbon capture & storage
- Reaching insignificant noise and vibration level
- Fuel cells with batteries are a promising option

DRIVER

- Environmental regulations & initiatives to
  - Zero-emission target for ship propulsion
  - Increase efficient of ship operation
  - Reduce Nox, Sox, CO2 and PM

WHY H2-PROPULSION?

- After emission limits for particulates, CO, HC and NOx have been tightened in the last years, this trend is expected to continue. In order to comply with the new standards, alternative propulsions systems as well as fuels should be considered. From this point of view, hydrogen is very promising, because it allows for a significant reduction of these emissions.
- Various ways of hydrogen production are developed / developing including innovative solutions of the local distribution of production sites. With special regard to the transport to the harbor, the potentials are elaborated and compared to current and predicted future requirements regarding a supply infrastructure. Currently, there are features of various hydrogen storage technologies, including storage in compressed gaseous and liquid form as well as Liquid Organic Hydrogen Carriers (LOHC) and metal hydride storages.

WHY H2-PROPULSION?

- Fuel Cells

- **The characteristics of relevant fuel cell types are introduced against the particularities and challenges regarding the usage of hydrogen in internal combustion engines.**
- **Fuel Cells:**
  - **Besides converting the chemical energy of the fuel to mechanical energy by means of combustions engines, it can also be converted to electrical energy in a fuel cell.**
  - **do not deliver mechanical energy as needed for propulsion, they have to be combined with an electric engine. Because electric motors require peak power only in a limited period of time, fuel cell systems are often decoupled from the propulsion engine by a buffer battery.**
  - **Due to the absence of high-temperature combustion, NOx emissions are minimized.**
  - **The oscillation-free operation reduces vibrations compared to internal combustion engines. This does of course not affect vibrations caused by the propeller.**

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WHY H2-PROPULSION?

- The oscillation-free operation reduces vibrations compared to internal combustion engines. This does of course not affect vibrations caused by the propeller.
- are expected to offer good reliability and availability
- A variety of fuel cell systems, but only the most promising candidates will be dealt with maritime applications. It appears that there is a preference for PEMFC and SOFC
- Fuel cell systems have a modular nature and their electrochemical performance tends to be independent of the system size
- Providers: Powercell (i.e. MS-100) with is advertised with an energy density 4-times as high as that of established competitors (see next page)
- For our application case, it was only considered to use fuel cells, which show best the zero-emission target.

WHY H2-PROPULSION?

Table 1: Gravimetric and volumetric specs for the fuel cell technologies discussed

TYPE OF FUEL CELL	Gravimetric Power Spec (kW/kg)	Volumetric Power Spec (kW/m3)	Efficiency (%)	Manufactur, Model
100 kW PEM	0.750	516	53	Powercell, MS-30
100/120 kW PEM	1.020	573	53	Powercell, MS-100
90 kW PEM	0.352	181	54	Ballard, Model FC Velocity HD
100 kW PEM	0.357	197	54	Ballard, Model FC Velocity HD
13.6 kW PEM	0.332	399	55	Nedstack, PemGen FCS 13-XXL
100 kW PEM	0.067	25	56	Nedstack, PemGen MT-FCPP-100
120 kW Power Rack	0.149	566	56	Hydrogenics Power Rack based on 4 HyPM HD 30 fuel cells
33kW PEM	0.421	407	55	Hydrogenics, Model HyPM HD 30

* Table 1 is based on different data sheets of providers

** In addition, a comparison of different fuel cells (providers) is shown in the reference, **Perspectives for the Use of Hydrogen as Fuel in Inland Shipping A Feasibility Study, October 2018**

WHY H2-PROPULSION?

- Internal combustion engines

- **Internal combustion engines:**
  - **using internal combustion engines (or ICE) seems beneficial for several reasons, especially applications requiring a high propulsion power and low energy consumption for auxiliaries can profit with a direct conversion.**
  - **The properties of hydrogen as fuel enable nearly zero emission combustion concepts. Emissions of carbon monoxide, hydrocarbons and particulates are fully oil borne if hydrogen is used as fuel. Compared to hydrocarbon fuels these emissions are initially very low.**
  - **The emission level for NOx emissions is comparable to a diesel or gasoline engine or even higher.**
  - **Estimated costs for ICE including gearbox and modifications for hydrogen operation (i.e. Total cost diesel fuel version / Total cost incl. modification for hydrogen operation = 120kEUR / 144kEUR for Rhine ferry (Engine Model C18 / 1x339kW)**

APPLICATION CASE

- Based on different papers and information as well as different studies. This is executed for a research vessel “FS Üthörn AWI”
  - Length 34.60m
  - Beam 9.00m
  - Draft 2.50m
  - Vmax 10kn

APPLICATION CASE

- **For this vessel with operating profile, two navigation conditions was studied:**
  - **Day trip: Vmax (10kn), 6h / V (4kn), 6h – ca. 3000kWh**
  - **Max. rang (5-days trip): Vmax (10kn), 120h – ca. 54000kWh**

    *** Including margin capacity than more 20%**
- **Information provided by power prediction and analyses revealed an average engine power of 400kW.**
- **Assuming the a same engine efficient with hydrogen operation, it is necessary 170 kg per day trip and ca. 3000kg per 5-days trip.**
- **Calculation of tank capacity for H2 and LOHC are shown in the following page, Table 2.**
- **For diesel fuel flow rates are between 200l/min and 900lmin, whilst for hydrogen fueling times for (5-days trip) are expected one hour.**
- **In addition, all calculations are shown on the Excel file:**
  - Berechnung Antriebskonzept-WIP05_200228_0.xlsx

APPLICATION CASE

Table 2: Calculation of tank capacity for LH2 and LOHC

		Day trip	Max.Range
Energy Shaft	kWh	3010	54173
Fuel Energy	kWh	5678	102213
Fuel Energy	MJ	20443	367966
Fuel Cell Efficiency (40 - 55%)	kWh	0.53	0.53
Hydrogen mass	kg	170	3067
LH2 tank volume	m3	2.41	43.33
LOHC tank volume	m3	3.00	54.04
40ft CH2 storage container***	number	0.06	1.04

* LH2 density 70.79kg/m3

** LOHC density 56.76kg/m3

*** Vol=52m2

* Table 2 is based on different data sheets of providers

** Tank volume is calculated based on the reference, **Perspectives for the Use of Hydrogen as Fuel in Inland Shipping A Feasibility Study, October 2018, page 80-82**

APPLICATION CASE

- **OPERATING COSTS:**
  - **In this study, for diesel fuel energy, it was considered 500EUR/t**
  - **For Hydrogen (LH2) fuel energy, it was considered 2500EUR/t, in the case of PEMFC + Hydrogen from Renewable Energies, and 2500EUR/t, in the case of PEMFC + Hydrogen from conventional methods.**
- **For example, the energy cost of the day trip condition were:**
  - **Diesel: ca. 330000EUR**
  - **LH2 + Renewables energies: ca. 470000EUR**
  - **LH2 + No Renewables energies: ca. 290000EUR**

APPLICATION CASE

- **A preliminary expected cost was calculated with the economic benefit to use a zero-emission fuel.**

Table 3: Expected cost of an alternative propulsion system with LH2 from Renewable Energies

Cost:	
CAPEX	1876000 EUR (high)
OPEX	180020 EUR
Economic benefit to society of emission reductions (30years)	414160 EUR

- **In addition, all calculations and explanations are shown on the Excel file:**
  - Berechnung Antriebskonzept-WIP05_200228_0.xlsx

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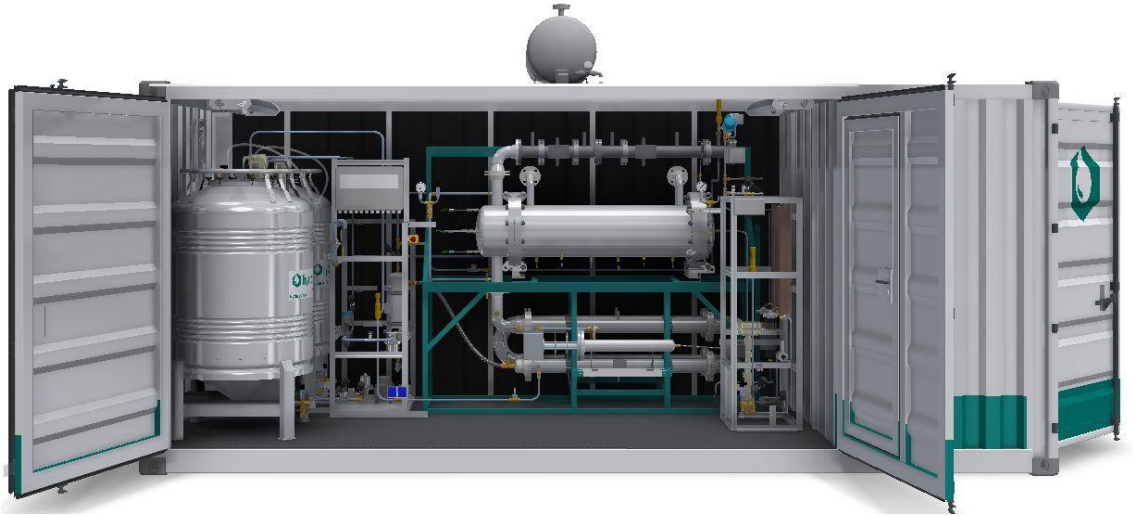
INFRASTRUCTURE

- Currently, non-commercial prototype installations, expected 2020
- Today, mainly industrial and land-based infrastructure
  - No demand for hydrogen fuel, really there is no distribution or bunkering in infrastructure for ships
  - but hydrogen fuel could be distributed in a similar manner as LNG
- Production of H2 from electrolysis is a known and available technology
  - It can be applied locally in port as well as an adequate supply of electrical power is available for the production process
- Liquid hydrogen (LH) might be transported to ports from storage sites where hydrogen is produced using surplus renewable energy (excess). Different solutions are coming
  - Standard 40ft containers for LH with a typical tank capacity of around (1000kg of hydrogen)

INFRASTRUCTURE

– **e.g. LOHC technology, which could be a perfect solution for a variety of applications in hydrogen logistics, as it enables the safe and easy hydrogen-handling by storing hydrogen in a liquid**.



* LOHC tank system of Hydrogenious LOHC Technologies GmbH, i.e. The hydrogen release rate of the Series 250 product, 22.5 kg/h, corresponds to a chemical power of the hydrogen of 750 kW.

ENVIRONMENTAL IMPACT

- Low-Emission alternative fuel for shipping, when hydrogen is generated from renewable or nuclear power using an efficient supply chain.
  - This is equal to the energy content of 150 Million tonnes of ship fuel
  - Nearly all hydrogen is produced from natural gas
- No limitations – Hydrogen produces from water using electrolysis (Clean Hydrogen). Moving towards Zero-Emission Transit
- Energy losses associated with hydrogen production from natural gas, i.e. Safety capturing & Storing the resulting CO2 (CCS)
- Fuel Cells (energy converters) does not produce any CO2 emissions (NO, NOx, SOx & Particulate Matter (PM) emission from ships), when using pure hydrogen (tank-to-propeller, TTP)
- Hydrogen fuelled internal combustion engines for marine applications (minimise GHG), but NOx emissions

COST

- Depends to a large extent on the electricity price (electrolysis) or gas/biogas (reformation)
- As a reference: a price of 70USD per Barrel is approx. 510USD/t fuel oil equivalent
- Electrolysis in areas with low electricity prices has the potential to drive the production costs down to between 3.5 & 4.1 USD/kg by 2020
- CAPEX
  - Similar as LNG fuelled engines (Piston engines for hydrogen)
  - Storage tanks for hydrogen are more expensive than LNG
  - For all other equipment, similar cost as in LNG systems
  - Fuel Cell technology is still under development
    - Cost: 3000 – 4000 USD/kW (expected 1000 USD/kW in 2022)

COST

- OPEX
  - Prices of hydrogen varies.
    - Fuel prices might depend on energy prices for hydrogen production.
    - Logistics costs might be considerably higher than LNG and other gases
  - Efficiency fuel to propeller higher than combustion engine
  - Less maintenance than conventional engines & turbines
  - Lifetime of fuel cells is shorter than that of piston engines or turbines.
  - Crew training requirements could be comparable to those of LNG / CNG.
  - Operational costs will be competitive when;
    - Same durability as combustion engine
    - Cost & time equal a general engine overhaul

COST

- **OTHERS**
  - **Cost of pollution rises significantly & strict emissions**

Batteries & Hydrogen are the future

CONCLUSION

- Hydrogen & batteries complement each other in an energy environment market by a growing renewable energy sector
  - Batteries to store relativity small amounts of energy for shorter duration
  - Hydrogen is better for long-term (e.g. seasonal) storage of large volumes of energy (e.g. using underground cavern).
- FCs mass production, beyond 2022, to allow production costs to reach a competitive level
- Only small maritime fuel cell applications with an electrical power output of up to 100kW are currently in operation
- Zero-emission target can be only obtain by using the hydrogen from renewable sources such as wind, solar, geothermal, or biomass.

REFERENCES

- Assessment of selected alternative fuels and technologies, DNV GL Maritime 2018-04
- Study on the use of Fuel Cells in Shipping, EMSA – DNV GL 2017
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- Hydrogen In The Maritime - The Old, The New And The Inevitable, Proton Motor Fuel Cell GmbH, Rohit Prasad
- Hydrogen-powered CSV moves zero emissions closer to reality, Riviera Magazine, John Snyder 2019
- Perspectives for the Use of Hydrogen as Fuel in Inland Shipping A Feasibility Study, October 2018. MariGreen, Maritime Innovations in Green Technologies